Omar Escobar

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EDUCATION

The University of Texas at San Antonio (UTSA) Expected Graduation: December 2027
Bachelor of Science in Mechanical Engineering, Minor in Computer Science

- Cumulative GPA 3.70 / 4.0 · Awards: President's List, Dean's List
- Relevant Coursework: Mechanics of Solids, Machine Element Design, Materials Engineering, Fluid Mechanics, Thermodynamics I & II, Heat Transfer, Numerical Methods, Measurements & Instrumentation, Dynamic Systems & Controls, Manufacturing Engineering

WORK EXPERIENCE

Naval Air Warfare Center Weapons Division, *Mechanical Engineering Intern, Virtual Institutes for Cyber and Electromagnetic Spectrum Research and Officer Training (VICEROY)*; China Lake, CA June 2026 – August 2026

- Designed components in SolidWorks for the U.S. Navy's only rocket sled test facility, a **21,550-foot** track rated to **Mach 4**; validated structure under high-acceleration loading through **finite element analysis** and delivered manufacturing-ready drawing packages with full geometric dimensioning and tolerancing.
- Coordinated design requirements directly with **NASA** and other government and industry stakeholders supporting test programs at the range.

UTSA Hypersonics Laboratory, *Mechanical Engineering Researcher*; San Antonio, TX August 2025 – August 2026

- Supported **hypersonic flow testing** in the Mach 7 wind tunnel, machining **test articles and fixtures** to tolerances required for supersonic and hypersonic test conditions and producing component diagrams for non-intrusive flow diagnostics.

National Science Foundation, *Mechanical Engineering Research Intern*; Gainesville, FL June 2025 – August 2025

- Developed and validated a data-driven method for **non-Gaussian turbulence characterization** in MATLAB, evaluating peak value distributions across homogeneous and machine-learning-generated heterogeneous terrain profiles.
- Quantified machine-learning-driven terrain roughness effects on turbulence using **Cobra probes** and published the resulting dataset to the national DesignSafe research repository under a permanent digital object identifier.

PROJECTS

Research Project – *MIT Lincoln Laboratory, U.S. Special Operations Command* September 2026 – Present
Autonomous Radio-Frequency Direction Finding

- Developing a **lightweight (5–10 lb) autonomous unmanned aircraft system** for the U.S. Special Operations Command Ignite 2027 challenge that navigates to a designated area, **passively detects and geolocates radio-frequency emitters**, and returns to launch unaided, using a **wing-integrated electronically steerable phased array antenna** and navigation without Global Positioning System signals.

Research Project – *MIT Lincoln Laboratory, U.S. Special Operations Command* September 2025 – August 2026
Under-Hull Uncrewed Underwater Vehicle Launch and Recovery System

- Developed an under-mounted unmanned underwater vehicle launch and recovery system, integrating hydrodynamic modeling, mechanical locking and release mechanisms, and sensor-triggered release.
- Optimized the system through computational flow simulation and data logging via embedded control units.

LEADERSHIP & COMMUNITY INVOLVEMENT

U.S. Army Reserve Officers' Training Corps – *Contracted Cadet, Simultaneous Membership Program; Battalion Signal Officer in Charge* May 2023 – Present

- **Air Assault School** graduate, ranked **top 5 of 32** cadets; Ranger Challenge competitor and Color Guard member; holds an **active Secret clearance**.
- Serves with the 4th Battalion, 133rd Field Artillery, Texas Army National Guard, San Marcos, TX.

UTSA First-Generation and Transfer Student Program – *Peer Mentor* August 2025 – Present

- Mentored **100+** students through their first year of engineering and received the program's **Rising Star Award**.

Additional Involvement: American Society of Mechanical Engineers (President); American Institute of Aeronautics and Astronautics (Vice President); Aeronautics and Rocket Club; Society of Hispanic Professional Engineers; Texas Society of Professional Engineers

SKILLS

Test and Instrumentation: Supersonic and hypersonic wind tunnel testing, multi-hole pressure probes, sensor integration, data acquisition and processing, strain measurement, materials characterization, rocket sled test support
Analysis and Modeling: Finite element analysis, ANSYS, computational flow simulation, hydrodynamic modeling, numerical methods, statistical characterization of non-Gaussian data
Design and Fabrication: SolidWorks, AutoCAD, geometric dimensioning and tolerancing, drawing package release, machine shop fabrication, 3D printing
Programming: MATLAB, Python, Arduino